



**Earn₹s**  
**DSA in C++**

# Pattern Printing and Functions

(Lecture 12)

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Summary Notes

# Contents

1. Need of Functions
2. Syntax of Functions
3. Arguments and Parameter
4. Inbuilt library Functions
5. Practice questions
6. Scope of Variable

# What **and** Why?

$$y = f(x) = x^2 + 3$$

$$g(x) = \sin x$$

output  
↑  
input

$$g\left(\frac{\pi}{2}\right) = 1$$

# The importance of Functions

In projects we always tend to use same / similar piece of code in the project multiple times, but not always we are looking for continuous repetition



Loop

# Syntax for Functions

## Wrapping the logic under a name

```
• void fun(){  
    ✓ cout<<"Tanvi is a obidient girl"<<endl;  
}  
✓ int main(){  
    ✓ cout<<"Hi Harry"<<endl;  
    ✓ fun(); // call lagai  
}
```

Output

• Hi Harry  
Tanvi is a . . . .  
.

## Output

. Hi Anu  
Hi Arjun  
Hi Sumit

```
✓void sumit(){  
|   ✓cout<<"Hi Sumit"<<endl;  
| }  
✓void arjun(){  
|   ✓cout<<"Hi Arjun"<<endl;  
|   ✓sumit();  
| }  
✓void anu(){  
|   ✓cout<<"Hi Anu"<<endl;  
|   ✓arjun();  
| }  
✓int main(){  
|   ✓anu();  
| }  
}
```

## Output

· Hi Anu  
· Hi Sumit  
Hi Arjun

```
• void sumit(){  
    ✓ cout<<"Hi Sumit"<<endl;  
}  
• void arjun(){  
    ✓ sumit();  
    ✓ cout<<"Hi Arjun"<<endl;  
}  
void anu(){  
    ✓ cout<<"Hi Anu"<<endl;  
    ✓ arjun();  
}  
✓ int main(){  
    ✓ anu();  
}
```

# Example

```
1  #include<iostream>
2  using namespace std;
3
4  void greeting() {
5      cout << "Good Morning" << endl;
6      cout << "Have a nice day" << endl;
7  }
8
9  int main() {
10     greeting(); // function calling
11     greeting();
12     greeting();
13 }
```

# Syntax for Functions

## The need for arguments

```
void minOfTwo(int a, int b){ // 2 numbers
    ✓ if (a < b) cout<<a<<" is smaller"<<endl;
    ✓ else cout<<b<<" is smaller"<<endl;
}

// isTareekeKoHumCameCaseKehteHai
• int main(){
    ✓ minOfTwo(71,34);
}
```

↓ ↓  
arguments

71 34  
a b

Output

34 is smaller

Q Write a f<sup>n</sup> which receives 3 numbers, and it prints their sum.

```
#include <iostream>
using namespace std;
void sum(int a, int b, int c){
    cout<<a+b+c<<endl;
}
int main(){
    sum(71,34,42);
}
```

# Badiya Example

Take a, b, c as input and print the following pattern: a = 3, b = 4, c = 5

```
*
* *
* * *
*
* *
* * *
* * * *
*
* *
* * *
* * * *
* * * * *
```

# Example

```
1  #include<iostream>
2  using namespace std;
3
4  void starTriangle(int x) {
5      for(int i = 1; i <= x; i++) {
6          for(int j = 1; j <= i; j++) {
7              cout << "*";
8          }
9          cout << endl;
10     }
11 }
12
13 int main() {
14     starTriangle(3);
15     starTriangle(4);
16     starTriangle(5);
17 }
```

Q//

|   | 1 | 2 | 3 | 4 | → i |
|---|---|---|---|---|-----|
| 1 | 1 | 1 | 1 | 1 |     |
| 2 | 1 | 2 | 2 | 2 |     |
| 3 | 1 | 2 | 3 | 3 |     |
| 4 | 1 | 2 | 3 | 4 |     |

n=4

i

```
for(i → 1 to n) {
    for(j → 1 to n) {
        if(i ≤ j) cout << i
        else cout << j
    }
}
```

C++

- Basics
- If-Else
- Switch
- ternary } α
- Loops
- PP → Only Lec-1
- Functions
- Pointers

Contest → Loops & PP

Monday → 6-8

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Contests



PW App

**Ques:** Print the given pattern

$n = 4$

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 2 | 2 | 2 | 2 | 2 | 1 |
| 1 | 2 | 3 | 3 | 3 | 2 | 1 |
| 1 | 2 | 3 | 4 | 3 | 2 | 1 |
| 1 | 2 | 3 | 3 | 3 | 2 | 1 |
| 1 | 2 | 2 | 2 | 2 | 2 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |

$n = 3$

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 |
| 1 | 2 | 2 | 2 | 1 |
| 1 | 2 | 3 | 2 | 1 |
| 1 | 2 | 2 | 2 | 1 |
| 1 | 1 | 1 | 1 | 1 |

$n = 2$

|   |   |   |
|---|---|---|
| 1 | 1 | 1 |
| 1 | 2 | 1 |
| 1 | 1 | 1 |

**Number Spiral**

**Ques:** Print the given pattern

1 2 3 4 3 2 1 → j

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 1 | 2 | 2 | 2 | 2 | 1 |
| 3 | 1 | 2 | 3 | 3 | 3 | 1 |
| 4 | 1 | 2 | 3 | 4 | 3 | 1 |
| 3 | 1 | 2 | 3 | 3 | 3 | 1 |
| 2 | 1 | 2 | 2 | 2 | 2 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |

n=4

Number Spiral

M-2 Ques: Print the given pattern

$$a+i == 2n$$

$$b+j == 2n$$

$$a = 2n - i$$

$$b = 2n - j$$

$$count \leftarrow \text{Min}(a, b)$$

|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | $\rightarrow j$ |
|---|---|---|---|---|---|---|---|-----------------|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |                 |
| 2 | 1 | 2 | 2 | 2 | 2 | 2 | 1 |                 |
| 3 | 1 | 2 | 3 | 3 | 3 | 2 | 1 |                 |
| 4 | 1 | 2 | 3 | 4 | 3 | 2 | 1 |                 |
| 5 | 1 | 2 | 3 | 3 | 3 | 2 | 1 |                 |
| 6 | 1 | 2 | 2 | 2 | 2 | 2 | 1 |                 |
| 7 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |                 |

$a$   $i$  Number Spiral